TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and pre sent potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclo sed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Gaurav Nanda	Email	coria-consortium@purdue.edu
Institute Name	Purdue University			
Address	401 N. Grant St., West Lafayette, IN 47907			

2. Technology Overview

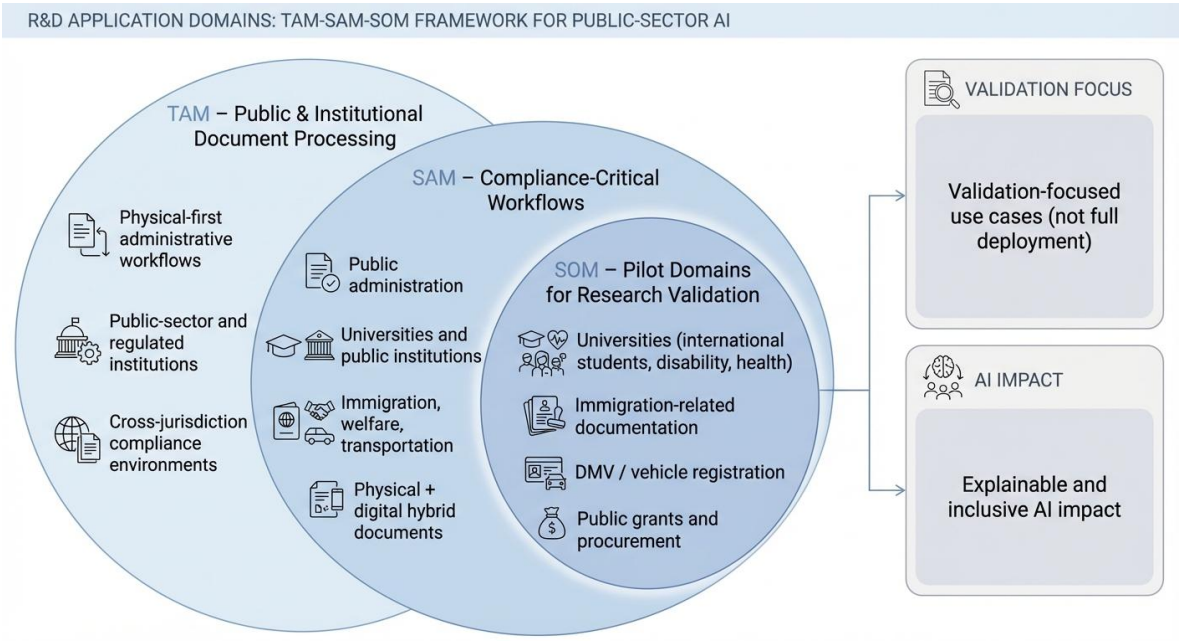
Title of Proposed Technology	**Physical-First Explainable AI for Public Document Compliance Across Jurisdictions**			
Korea's 12 major national strategic technology fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (Major Category)**		**Technology Field 2 (Major Category)**	
	Information & Communication		***Knowledge Services***	
	Medium category	Small category	Medium category	Small category
	Software	***Internet SW***	***R&D / Engineering Services***	***Testing / Inspection / Analysis Techniques***
Proposed technology concept	**Core Concept** The proposed technology is an integrated physical-first AI system for public document processing, designed to support trustworthy and explainable decision-making in public administration. The core concept is to bridge the gap between physical-origin documents (e.g., handwritten forms, signatures, scanned paper documents) and digitally auditable administrative processes, while maintaining transparency, accountability, and human oversight. **Key Functions and Potential Applications** **Key Functions** 1. Physical Document Capture and Understanding ■ Acquisition and interpretation of documents originating from physical media, including			

handwriting, annotations, checkboxes, and signatures.

- Robust processing under realistic capture conditions such as noise, skew, and partial occlusion.

2. Explainable Compliance Reasoning
   - Application of jurisdiction-specific administrative rules to extracted document content.
   - Generation of traceable reasoning outputs that link decisions to both physical document regions and applied rules.
   - Provision of uncertainty and confidence information to support review and appeal processes.
3. Agentic Workflow Support with Human Oversight
   - Assistance in document routing, validation, escalation, and correction through agent-based workflows.
   - Explicit integration of human-in-the-loop decision points to ensure accountability and prevent full automation of high-risk decisions.
4. Cross-Jurisdiction Adaptability
   - Modular handling of regulatory differences across institutions, regions, and countries with multilingual inclusive ai support.

Market Context



The global intelligent document processing (IDP) market is growing from $1.8B (2024) to $10.2B (2030), with the government sector expanding at 31.2% CAGR. Despite digitalization efforts, 78% of government documents still originate in paper form (Gartner 2024).

Korea faces urgent needs: ₩2.3T AI automation investment (Digital Government Innovation Plan 2024-2027), Korea Digital Act mandating AI explainability by 2026, and 17 ministries + 243 local governments seeking solutions for 12M+ annual paper documents. The US shows similar demand: USCIS processing 8M+ applications (65% handwritten), 250M+ DMV transactions, and $1.2B federal

	budget for trustworthy government AI. Current solutions fail: Legacy OCR achieves only 60-70% accuracy, commercial IDP lacks explainability, and academic research remains at TRL 2-3. Societal urgency is critical—30% of Korean elderly are excluded from digital services, USCIS processing times have increased 250%, and 68% of the US public distrusts government AI. **Technical Innovations** ● Physical-First Document Understanding: Novel sensor fusion combining scanning + depth sensing for 3D form analysis, adversarial training on non-standard handwriting (elderly, disabled, non-native), uncertainty estimation at character level for selective human review ● Explainable Compliance Reasoning: Hybrid symbolic-neural approach linking physical regions → extracted fields → compliance decisions, structured reasoning logs with bidirectional traceability, confidence propagation through reasoning chain ● Agentic Workflow: Multi-agent system with confidence-aware decision logic, adaptive escalation based on workload and risk profile ● Cross-Jurisdiction Adaptability: Modular rule engine with jurisdiction-specific plugins, ontology-based representation, transfer learning for rapid adaptation **Potential Applications** Potential Applications: Immigration/visa processing, vehicle registration, welfare administration, customs declarations, university administrative workflows, and broader trustworthy AI adoption in public services where physical documents remain prevalent.
Rationale for Joint R&D Project	**Critical Technical Barriers** Developing a physical-first explainable AI system for public document compliance requires overcoming several fundamental technical barriers that extend beyond the capabilities of a single organization or country. First, physical document understanding in government contexts involves highly variable handwritten forms, signatures, stamps, annotations, and multilingual content captured under suboptimal conditions. Achieving robust interpretation requires advances in multimodal perception, noise-resilient representation learning, and alignment between physical document regions and digital semantics. Second, explainable regulatory reasoning across jurisdictions presents a major challenge: legal and administrative rules vary between ministries, municipalities, and countries, and must be encoded into interpretable computational formats. Ensuring that AI outputs remain transparent, auditable, and legally defensible requires integrating rule-based systems, natural language reasoning, and uncertainty modeling that remain technically immature. Third, high-stakes administrative workflows demand human-centered oversight, but current AI systems lack mechanisms to provide traceable rationales, confidence measures, and human-AI collaborative checkpoints. Designing workflows that maintain accountability while enabling automation requires breakthroughs in explainable AI interfaces, cognitive load modeling, and validation protocols.

(max 10 pages)

	Why Global Collaboration? These challenges necessitate a joint R&D framework combining world-class academic expertise with real-world public sector experience, and Purdue University provides suitable capabilities for this effort. Purdue's research ecosystem includes the Purdue Institute for Physical Artificial Intelligence (IPAI), which leads advancements in AI grounded in physical sensing and multimodal document interpretation; and CERIAS, one of the world's foremost centers for trustworthy and accountable computing. These institutions offer deep strengths in explainable AI, cross-jurisdiction reasoning, multilingual document processing, and rigorous evaluation frameworks that have capabilities essential for building credible AI systems for government use. Beyond these flagship institutes, Purdue maintains a broad ecosystem of AI-focused research groups across computer science, human-centered computing, linguistics, and information systems—covering areas such as explainable AI, multilingual NLP, human-AI interaction, and formal verification. Through global collaboration, the project can leverage Purdue's foundational research alongside practical administrative knowledge and diverse international workflows, enabling robust cross-regional validation, accelerating the development, and establishing the trustworthiness required for deployment in high-stakes public-sector environments.
Key words	1. Physical Document Understanding & Document Intelligence 2. Explainable and Trustworthy AI 3. Human-Centered AI Systems 4. Regulatory Reasoning Frameworks 5. Cross-Jurisdiction Validation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2]	**Start of the Project**	Level (2)	**End of the Project**	Level (5)
R&D Objectives	**R&D Objectives** The primary objective of this research project is to design, implement, and empirically validate a unified framework for physical-first, explainable, and adaptive AI-based public document compliance. **Detailed Objectives** 1. Physical-First Document Understanding Develop methods for robust interpretation of public documents originating in the physical world, including handwritten text, checkboxes, annotations, and physical signatures, with particular attention to non-standard and assisted handwriting produced by elderly, disabled, or non-native users. 2. Explainable and Auditable Compliance Reasoning Establish explainability mechanisms that enable traceability from physical document inputs to extracted fields and compliance decisions, ensuring that all automated reasoning processes are transparent, contestable, and suitable for audit and appeal in public-sector			

	contexts. 3. Agentic and Adaptive Workflow Modeling Investigate agent-based approaches for public document workflows that support adaptive routing, validation, and escalation under jurisdiction-specific regulatory constraints, while preserving human oversight and accountability. 4. Cross-Jurisdiction Generalization Evaluate the generality of the proposed framework across multiple jurisdictions and administrative domains, demonstrating that the underlying methods are not limited to a single regulatory system or document type.
R&D Target Specifications	1. Physical-First Document Understanding The system is expected to reliably interpret documents that originate in physical form, including handwritten and manually annotated public documents. Handwritten text recognition performance will be evaluated at the character level, with a target accuracy of at least 85% on representative public-sector forms. The evaluation set will explicitly include non-standard and assisted handwriting samples, reflecting documents produced by elderly users, persons with disabilities, and non-native writers. For predefined key fields (such as personal identifiers, dates, and selection fields), the target field extraction accuracy is 90% or higher, measured against manually verified ground-truth annotations. In addition, the system should correctly identify the structural components of public forms such as text fields, checkboxes, and signature regions with an accuracy of approximately 90%. Robustness will be assessed by introducing common physical artifacts (e.g., scanning noise, skew, partial occlusion). Under these conditions, performance degradation should remain within 10% relative to a clean-scan baseline. 2. Explainability and Auditability A core requirement of the system is that all automated outputs remain transparent and suitable for review in public-sector contexts. All extracted fields are required to be traceable to their physical origins. Accordingly, all extracted fields (100%) must be linked to specific source regions in the document image or sensor input. For compliance-related decisions, the system should generate structured reasoning records that document which rules were applied and how intermediate conclusions were reached. The target is that all automated compliance decisions are accompanied by such reasoning logs. The system should also provide confidence or uncertainty estimates for extracted fields and final decisions. These estimates should be generated consistently for all outputs, enabling reviewers to distinguish between high-confidence and ambiguous cases. Finally, any human intervention such as edits, overrides, or escalation must be recorded with appropriate provenance information (e.g., timestamps and action types), ensuring full

traceability during audits or appeals.

3. Agentic Workflow Performance

The agentic workflow component is intended to support, rather than replace, human decision-making. Its effectiveness will therefore be evaluated primarily in terms of workload reduction and appropriate escalation.

In pilot scenarios, the system should achieve a reduction of at least 30% in routine manual review effort compared to baseline manual processing.

For cases involving ambiguity or low confidence, the system should reliably escalate decisions to human reviewers. The target recall for correctly identifying such cases is 95% or higher, prioritizing safety over automation.

When simulated regulatory updates are introduced, the system should maintain stable performance. In particular, no statistically significant increase in error rates should be observed following rule updates ($p < 0.05$), relative to pre-update performance.

4. Cross-Jurisdiction and Multilingual Robustness

To demonstrate generality, the system will be evaluated across at least two jurisdictions with distinct regulatory frameworks.

Performance will also be assessed across multiple administrative domains, with successful evaluation on a minimum of three public-sector document types (e.g., immigration, vehicle registration, welfare-related documents).

Multilingual capability will be validated on documents in at least two languages. For secondary languages, extraction accuracy degradation should remain within 10% of the primary-language baseline.

5. System Integration and Validation Readiness

The integrated system should be capable of processing all evaluation documents end to end, from physical ingestion through compliance reasoning and explainable output generation. All test documents should complete this pipeline without silent failures.

Successful completion of these targets will support validation of the research outcomes at Technology Readiness Level 6, demonstrating that the proposed technologies function reliably in relevant, controlled environments.

Capability	Legacy OCR	Commercial IDP	Proposed Technology
Handwriting accuracy	60-70%	75-82%	85%+ (real-world)
Explainability	None	Rule logs	Physical-traceable reasoning
Regulatory reasoning	None	Templates	Adaptive compliance
Cross-jurisdiction	Single country	Manual reconfig	Modular framework
Human oversight	Manual review	Escalation rules	Confidence-aware agentic

R&D Plan

Research Approach

This project adopts a phased and implementation-aware research approach that explicitly accounts for the development of physical document capture and processing systems in addition to

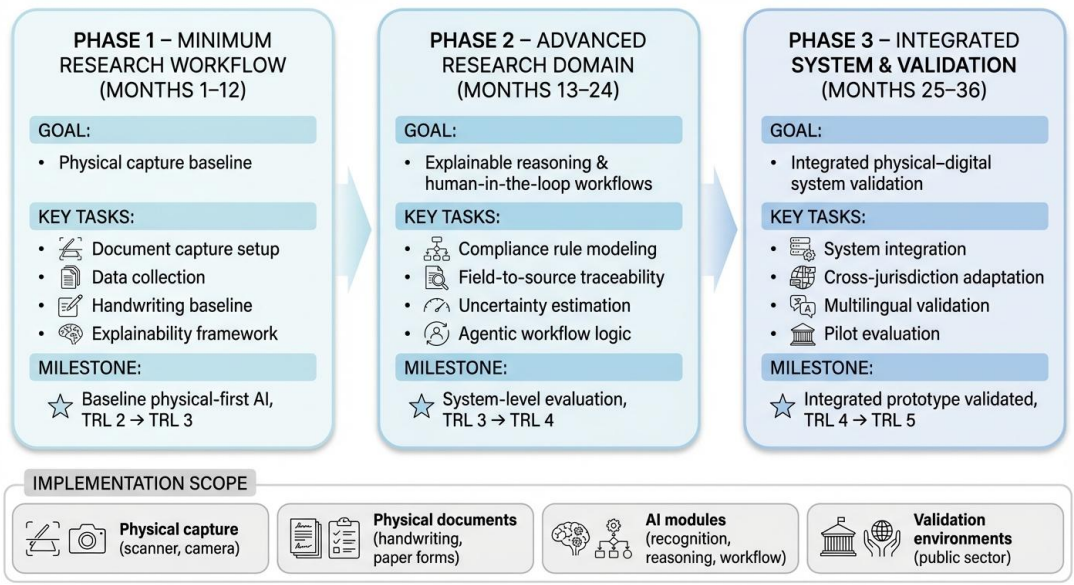
algorithmic research. Because the target application involves documents originating in the physical world, the research considers practical implementation constraints—such as sensing, capture quality, and deployment environments—from the earliest stages. Core technologies are developed modularly and integrated through iterative prototyping and evaluation.

Key Tasks and Methodologies

1. Physical Document Capture and Understanding
   - Design and evaluate capture pipelines using scanners and camera-based devices
   - Collect and annotate physical public documents reflecting real usage conditions
   - Develop handwriting and layout analysis models robust to capture variability
   - Methodologies: Sensor-based data acquisition, multimodal document analysis, controlled capture experiments
2. Explainable Compliance Reasoning
   - Formal modeling of compliance rules linked to extracted physical fields
   - Development of traceability mechanisms connecting decisions to physical document regions
   - Uncertainty estimation and audit-log generation for public-sector review
   - Methodologies: Knowledge-based rule modeling, explainable AI techniques, provenance tracking
3. Agentic Workflow and Human-in-the-Loop Integration
   - Modeling of workflows including human review, correction, and escalation
   - Design of confidence-aware decision logic for operational settings
   - Evaluation of workflow efficiency and decision stability
   - Methodologies: Agent-based workflow modeling, human-in-the-loop experimentation
4. System Integration and Cross-Jurisdiction Validation
   - Integration of physical capture components with AI reasoning modules
   - Adaptation to multiple regulatory contexts and languages
   - End-to-end evaluation in controlled pilot environments
   - Methodologies: Modular system integration, comparative pilot studies

(max 10 pages)

Timeline and Milestones



Risk Management

Risk	Probability	Impact	Mitigation
Handwriting <85% accuracy	Medium	High	Ensemble models, expand training data, adjust to 82% if needed
Cross-jurisdiction transfer fails	Low	High	Manual configuration tool backup, extend pilot +3 months
Regulatory changes mid-project	High	Medium	Quarterly regulatory scanning, modular design for rapid updates
Korea-US communication barriers	Medium	Medium	Dedicated PM, overlap hours, monthly in-person sync
Insufficient document samples	Medium	High	Partner with 3 gov agencies, synthetic data generation
Pilot not representative	Low	High	Multi-site validation (immigration + DMV + welfare)

Contingency Plans:

- If Phase 1 delays >2 months → Compress Phase 2, reduce scope to 2 jurisdictions
- If commercial pilot fails → Pivot to academic validation with extended publication focus

(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	Purdue University's Roles and Responsibilities (R&R) ● Foundational Research (Months 1-24): Develop theoretical frameworks for physical-first XAI, design uncertainty quantification methods, create formal models for regulatory reasoning ● Algorithm Development (Months 6-30): Implement handwriting recognition models, develop confidence-aware decision logic, design cross-jurisdiction transfer learning, research code ● Evaluation and Validation (Months 18-36): Establish benchmark datasets, conduct controlled experiments, perform user studies with government administrators, and lead academic writing ● Knowledge Transfer (Throughout): Host Korean researchers, monthly video conferences, quarterly in-person workshops, co-supervise Korean students
R&R of Korean SME partners	Korean SME's Role and Responsibilities (R&R) ● System Development (Months 1-36): Design physical capture pipeline, integrate Purdue algorithms into production, develop web admin interface, build agentic workflow orchestration, deploy AWS infrastructure ● Data Collection (Months 1-18): Partner with 3+ government agencies, collect 10,000+ physical document samples, annotate ground-truth data, ensure privacy compliance (PIPA) ● Pilot Deployment (Months 24-36): Deploy prototype at 3 government sites, 6-month field validation with administrative staff, gather user feedback, measure system performance, produce technical documentation ● Validation & Test (Months 30-36+): Perform end-to-end system validation, including functional testing, evaluation, and workflow verification.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Technical Impact ■ By validating across both Korean (high-volume workflows) and US (diverse regulatory regimes) contexts, the project produces internationally applicable technology. ■ Development of an explainable compliance reasoning engine linking administrative decisions to both physical document regions and jurisdiction-specific rules, enabling auditable and trustworthy AI behavior. ■ Advancement of uncertainty-aware human-in-the-loop workflows, improving stability, decision traceability, and operational safety in government settings. ● Economic & Industrial Impact ■ Significant reduction in document processing time and administrative workload, enabling governments to reallocate labor toward higher-value services. ■ Foundation for a scalable public-sector AI platform, opening commercialization opportunities in immigration, welfare, licensing, vehicle registration, and education

(max 10 pages)

	administration. ■ Strengthening of the AI compliance technology industry, enabling domestic companies to compete in global IDP (Intelligent Document Processing) and public-sector AI markets. ● Social & Environmental Impact ■ Improved accessibility of public services for elderly, underserved, and digitally marginalized populations by enabling more accurate and reliable processing of physical documents still widely used in government workflows. ■ Increased public trust in government AI through transparent decision-making, explainability, and independent academic validation. ■ Enhanced fairness and accountability in administrative decision processes by reducing human errors and supporting consistent, rule-based outcomes. ■ Reduction in paper waste and environmental burden through digitization pipelines that minimize rescanning, reprocessing, and redundant physical handling.

6. References